

Statistical Methods

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Outline I

- 1 Course Overview
- 2 Data Summarization
- 3 Populations, Samples, and Experiments
- 4 Bivariate Data: Covariance and Correlation

Course Description & Objectives

- Provides a **rigorous foundation** in statistical theory, probability modeling, and computational data analysis.
- Teaches data summarization, visualization, and application of standard probability models.
- Utilizes **Monte Carlo simulations** to illustrate sampling distributions.
- Covers parameter estimation, hypothesis testing, linear/logistic regression, and time series analysis.
- Integrates theoretical concepts with modern computational techniques like **Bootstrap resampling in Python**.

Grading Policy

Evaluation will follow a **Relative Grading Policy**:

Assessment Component	Weightage
Lab Works & Assignments	20%
In-Semester Exam I	20%
In-Semester Exam II	20%
End-Semester Examination	40%

Table: Course Weightage Breakdown

Course Materials & References

- **Sheldon Ross**, *Probability and Statistics for Engineers and Scientists*.
- **William Feller**, *An Introduction to Probability Theory and Its Applications (Vol. 1)*.
- **George Casella & Roger L. Berger**, *Statistical Inference*.
- **Bradley Efron & Robert J. Tibshirani**, *An Introduction to the Bootstrap*.

What is Data?

- **Data** refers to a collection of facts, observations, measurements, or descriptions gathered for analysis.
- It serves as the raw material from which we extract meaningful information, patterns, and insights.
- **Dataset:** A structured collection of data, typically organized in rows (representing individual observations or records) and columns (representing variables or features).

Example

The heights of all students in the class, their undergraduate backgrounds, and their exam scores all constitute data.

Broad Classification of Data Types

Before analyzing data, we must understand its type, as this determines which statistical methods and visualizations are appropriate. Data is broadly classified into two main categories:

- 1 **Qualitative (Categorical) Data:** Describes categories, attributes, or labels without a natural numerical meaning.
- 2 **Quantitative (Numerical) Data:** Represents measurable quantities or discrete counts. It inherently answers questions like "how much" or "how many."

1. Qualitative (Categorical) Data

Nominal Data

Categories with **no inherent order or ranking**. The labels are simply used to distinguish different groups.

- *Examples:* Blood type (A, B, AB, O), pin codes, or preferred programming language (Python, R, Julia).

Ordinal Data

Categories that have a **meaningful order or ranking**, but the mathematical distance between categories is not standardized.

- *Examples:* Customer satisfaction ratings (Poor, Fair, Good, Excellent), or letter grades (A, B, C, D, F).

2. Quantitative (Numerical) Data

Discrete Data

Countable values that represent separate, distinct units. These are typically whole numbers and cannot logically be broken down into fractions.

- *Examples:* Number of students attending a lecture, the number of errors in a process, or the number of cars passing through a street in an hour.

Continuous Data

Measurable values that can take **any** numeric value within a given range, including infinite possibilities of fractions and decimals.

- *Examples:* A student's exact height, ambient temperature, or humidity.

A Real-World Scenario: Raw Data

Imagine you surveyed 25 incoming Data Science students about their primary programming language.

You collect the following raw data (a nominal categorical dataset):

Raw Survey Results

Python, R, Python, C++, Java, Python, R, R, Python, Python, Java, C++, Python, R, Python, Python, R, Java, C++, Python, R, Python, Python, Java, R

The Challenge

The Situation: The Head of the Department needs to plan the software installations for the lab. They ask you two questions:

- 1 Which language is the most popular?
- 2 What exact percentage of the class prefers R?

Question for the Class

Staring at that raw list of 25 words is not helpful.

How would you reorganize or present this data so the Head of Department can instantly see the answers they need?

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The Solution: The Frequency Table

The most logical way to represent this data is by creating a **Frequency Table**. We group identical responses and tally them up.

Programming Language	Number of Students (Frequency)
Python	11
R	7
Java	4
C++	3
Total (n)	25

Table: Absolute Frequency of Preferred Languages

Now, question 1 is instantly answered: Python is the most popular.

Enhancing the Table: Answering Question 2

To answer the second question ("What percentage prefers R?"), we need to expand our table to include **Relative Frequency**.

Language	Frequency (f)	Relative Freq (f/n)	Percentage
Python	11	$11/25 = 0.44$	44%
R	7	$7/25 = 0.28$	28%
Java	4	$4/25 = 0.16$	16%
C++	3	$3/25 = 0.12$	12%
Total	$n = 25$	1.00	100%

Question 2 answered: Exactly 28% of students prefer R.

Formalizing the Concepts & Terminology

Let's formally define what we just did. For a dataset of size n :

- **Absolute Frequency (f):** The actual number of times a specific value or category appears in a dataset.
- **Relative Frequency:** The proportion of the total dataset that a specific value represents.

$$\text{Relative Frequency} = \frac{f}{n}$$

- **Percentage:** The relative frequency multiplied by 100.

Formal Table Nomenclature

- **Frequency Distribution Table:** The formal name for a table displaying the absolute frequencies of different categories.
- **Relative Frequency Distribution Table:** The formal name when the table is expanded to include proportions or percentages.

Moving from Tables to Visuals

We have our Frequency Distribution Table, but scanning numbers is not always the fastest way for the human brain to process information.

Why Visualize?

Visualizations help us instantly spot trends, majorities, minorities, and outliers without having to analyze individual numbers. They make data **actionable**.

For **Categorical Data** (like our nominal programming languages), the two most common and effective visual tools are:

- 1 Bar Charts (Bar Graphs)
- 2 Pie Charts

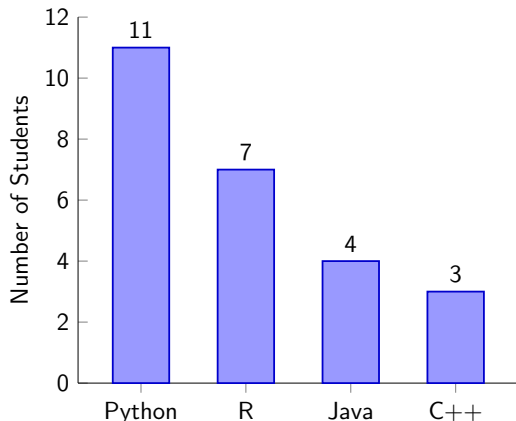
1. The Bar Chart: Visualizing Frequency

A **Bar Chart** uses rectangular bars where the height is strictly proportional to the frequency of the category.

Mapping our data:

- **X-axis:** Distinct categories (Languages).
- **Y-axis:** Absolute frequency (Number of students).

Notice how the height instantly communicates Python's popularity.

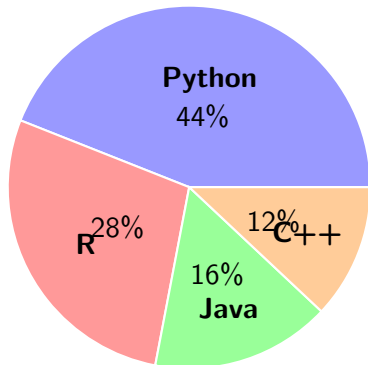


2. The Pie Chart: Visualizing Proportions

A **Pie Chart** represents data as slices of a circle. The entire circle represents 100% of the data, and each slice is proportional to the **relative frequency**.

Mapping our data:

- The whole pie represents all $n = 25$ students.
- **Python**: 44% of the circle.
- **R**: 28% of the circle.

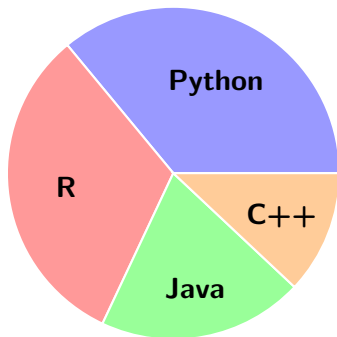


Note: Statisticians often prefer bar charts, as the human eye compares heights better than angles!

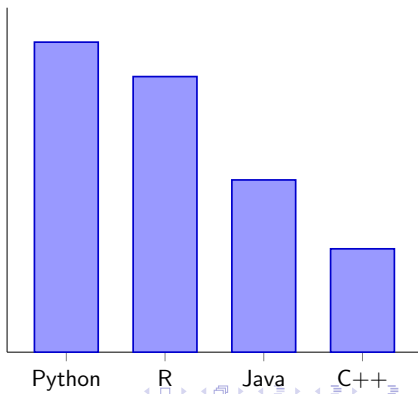
Why Bar Charts are usually preferred

Let's slightly change our scenario so that **Python** and **R** have very similar frequencies. I have removed the numbers. **Can you tell which language is more popular just by looking?**

Pie Chart



Bar Chart



Visualizing Discrete Ungrouped Data

Before we group data into bins (like in a histogram), we often need to look at raw, **discrete data**.

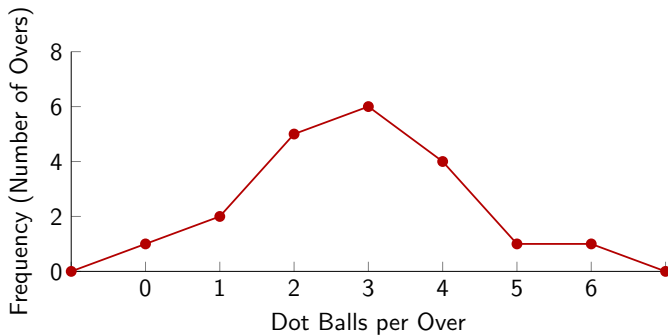
For this type of univariate data, we want visualizations that show the frequency distribution **without losing the original values**. The two best tools for this are:

- 1 Frequency Polygons
- 2 Stem-and-Leaf Displays

Frequency Polygons

A **Frequency Polygon** is formed by taking the points from a line graph and connecting successive points with straight lines to emphasize the shape and trend of the distribution.

Example: Number of dot balls per over in a T20 cricket innings (20 overs).



2. Stem-and-Leaf Display

When discrete data has wider variance (e.g., test scores from 0 to 100), drawing a dot plot becomes impractical. A **Stem-and-Leaf Display** solves this by splitting the numbers.

How it works:

- **Stem:** The leading digit(s) (e.g., the tens place).
- **Leaf:** The final trailing digit (e.g., the ones place).

Dataset (Exam Scores):

45, 52, 56, 60, 63, 63, 68,
71, 74, 75, 75, 79, 82, 88, 91

Stem	Leaf
4	5
5	2 6
6	0 3 3 8
7	1 4 5 5 9
8	2 8
9	1

Why use this?

If you rotate the table to the left, this table looks exactly like a Bar Chart! It provides a visual distribution shape while preserving the raw data.

Key: 7|1 means 71

Summary: Choosing the Right Visual Tool

We now have a toolkit for handling different types of simple, 1D (univariate) data before moving into more complex grouped or bivariate analysis.

Data Type	Best Visual Tool	Key Feature
Categorical (Nominal)	Bar Chart	Compares category totals
Categorical (Proportions)	Pie Chart	Shows parts of a whole
Discrete (Small Range)	Frequency Polygon	Shows exact discrete counts
Discrete (Moderate Range)	Stem-and-Leaf	Shows distribution & raw value

Moving to Continuous Data

As part of our study on data summarization, we must transition from discrete counts to **continuous data** (e.g., exact execution times, temperatures, or precise weights).

The Challenge

Unlike discrete data (where values like 0, 1, or 2 repeat frequently), continuous data often yields unique, highly specific values.

Example: If some one measures the temperature during different times of the day, their exact readings may be 32.4, 35.1, 33.9, 31.2, 32.5 — meaning almost every value has a frequency of 1.

A standard frequency table or line graph of raw continuous data is completely flat and uninformative because nothing repeats!

The Solution: Binning (Class Intervals)

To analyze continuous data, we must group individual observations into contiguous, non-overlapping intervals called **bins** or **class intervals**.

- **Binning (Grouping):** Converts continuous data into aggregated frequency counts per interval.
- **Class Boundaries:** The upper and lower limits of each bin (e.g., 50 to < 60 , 60 to < 70).
- **Bin Width:** The uniform size of each interval.

The Binning Trade-off

- **Too few bins:** Oversimplifies the data, losing structural details.
- **Too many bins:** Retains too much noise, preventing clear pattern recognition.

The Histogram

Once continuous data is binned, the primary visual tool used in data visualization is the **Histogram**.

Definition

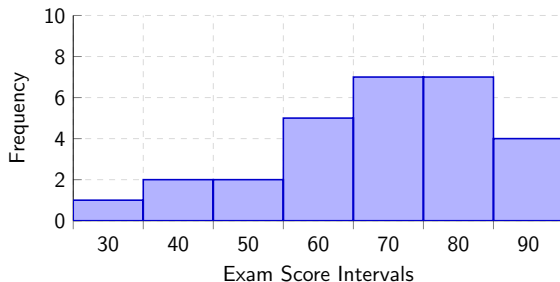
A histogram is a graphical display of binned quantitative data where the frequency of each interval is represented by the size of a vertical bar.

Key Differences from a Bar Chart:

- **No Gaps:** Because the underlying data is continuous, the bars **touch each other** (unlike bar charts which separate distinct categories).
- **X-axis:** Represents continuous numerical scales (class boundaries), not discrete names.

Example: Histogram

Scenario: Exam scores of 30 students grouped into bins of 10 points.



The touching bars emphasize the continuous numerical transitions across intervals.

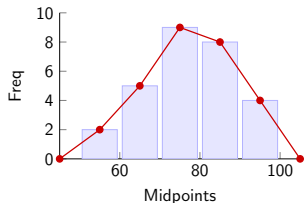
Note: *Though exam scores are discrete in nature if they are considered to be whole numbers their range often makes it desirable to use histograms instead of other tools.*

Frequency Polygons for Histograms

We can transform or overlay a histogram with a **Frequency Polygon** (a specialized line chart) to clearly track distribution shapes.

Construction Steps:

- 1 Determine the **midpoint** of each histogram bin.
- 2 Plot the frequency at each midpoint.
- 3 Connect points with straight line segments.
- 4 Anchor ends to the X-axis.



The frequency polygon provides a clean line-graph outline showing the overall trend of the continuous dataset.

Moving from Univariate to Bivariate Data

So far, we have focused on **univariate data**—analyzing a single variable at a time (e.g., just exam scores, or just heights).

What is Bivariate Data?

Bivariate data involves the simultaneous measurement of **two different variables** (X and Y) on each individual or observation in a dataset.

Examples:

- X = Study hours, Y = Exam score.
- X = Temperature, Y = Ice cream sales.
- X = Advertising budget, Y = Product revenue.

The Purpose of Bivariate Analysis

When working with bivariate data, our primary goal shifts from summarizing a single distribution to investigating the **relationship or association** between the two variables.

- **Core Question:** Do changes in variable X relate to predictable changes in variable Y ?
- **Dependency:** We examine whether Y depends on X —laying the fundamental groundwork for **Regression Analysis**.

The Core Visual Tool

To visualize the relationship between two continuous quantitative variables, we use a **Scatter Plot**.

The Scatter Plot

A **Scatter Plot** is a two-dimensional graphical diagram where each observation is represented as a single point (X_i, Y_i) mapped onto a Cartesian coordinate system.

- **Horizontal Axis (X-axis):** Represents the independent or explanatory variable.
- **Vertical Axis (Y-axis):** Represents the dependent or response variable.

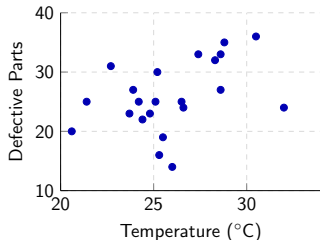
Key Patterns to Look For:

- 1 **Direction:** Positive trend (variables move together) or negative trend (move in opposite directions).
- 2 **Form:** Linear versus non-linear patterns.
- 3 **Strength:** How tightly clustered the points are around a trend.

Example: Scatter Plot (Temperature vs. Defective Parts)

Problem Statement:

- Analyzing the relationship between daily midday temperature ($^{\circ}\text{C}$) and the number of defective parts over 22 workdays.
- Variables:**
 - X: Temperature ($^{\circ}\text{C}$)
 - Y: Number of Defective Parts



The scatter plot maps all 22 bivariate data pairs to check for underlying patterns, spread, or correlation.

Time Series Data

Having explored univariate and bivariate data, we now turn to a specialized and critical form of sequential data: **Time Series Data**.

What is Time Series Data?

A sequence of data points recorded at regular, successive intervals of time (e.g., hourly, daily, monthly, or yearly).

Examples:

- Daily closing stock prices valuations.
- Monthly rainfall measurements or average temperatures over a decade.
- Annual company revenues or Gross Domestic Product (GDP) growth rates.

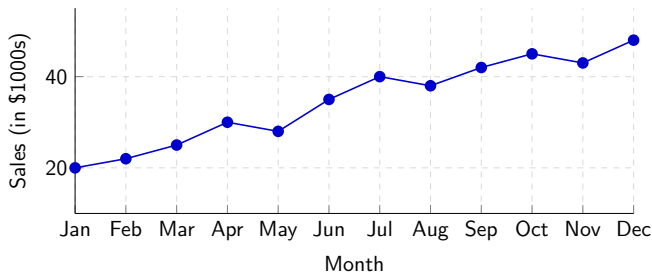
Line Charts for Time Series Data

When visualizing time series data, the **Line Chart** (or time-series plot) is the absolute standard and most effective tool.

- **Horizontal Axis (X-axis):** Represents time in strict chronological order.
- **Vertical Axis (Y-axis):** Represents the variable value being tracked.
- **Connecting Lines:** Emphasize trajectory, temporal trends, seasonal fluctuations, and long-term direction over time.

Example: Monthly Sales Time Series

Scenario: Tracking a company's monthly product sales across a single year.



The upward trajectory clearly reveals a positive growth trend over the course of the year.

Numerical Summaries: Descriptive Measures

While data visualizations help us see patterns, we need precise **numerical measures** to quantify key characteristics of a dataset.

The Four Pillars of Descriptive Statistics

- ➊ **Central Tendency:** Where is the center or typical value of the data?
- ➋ **Dispersion (Spread):** How much do the data values vary or spread out?
- ➌ **Shape:** Is the distribution symmetric, skewed, or peaked?
- ➍ **Association:** How strongly do two variables move together?

Let's explore each category in detail.

1. Measures of Central Tendency

Central tendency single values represent the middle or typical point of a dataset.

- **Sample Mean ($\bar{x}$):** The arithmetic average, balancing all values.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

- **Median (M):** The exact middle value when data is sorted in ascending order. Robust against extreme outliers.
- **Mode:** The most frequently occurring value in the dataset (essential for categorical and discrete data).

Choosing the Right Center

If a dataset has heavy outliers (like income or housing prices), the **median** is often preferred over the mean because it isn't skewed by extreme values.

2. Measures of Dispersion (Spread)

Knowing the center isn't enough; we must measure how dispersed the data points are around that center.

- **Mean Absolute Deviation (MAD):** The average distance between each data point and the mean.

$$\text{MAD} = \frac{\sum |x_i - \bar{x}|}{n}$$

- **Sample Standard Deviation (s):** The most common measure of spread, representing the average distance from the mean using squared units.

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Squaring differences in standard deviation heavily penalizes large deviations, making it mathematically optimal for statistical inference.

3. Measures of Shape: Skewness and Kurtosis

Beyond center and spread, we describe the geometric shape of a distribution curve.

Skewness (Asymmetry):

- **Symmetric:** Mean = Median.
- **Right-skewed (Positive):**
Long tail stretches to the right
(Mean > Median).
- **Left-skewed (Negative):**
Long tail stretches to the left
(Mean < Median).

Kurtosis (Tailedness):

- Measures the "peakedness" or heavy tails of a distribution relative to a normal distribution.
- **Leptokurtic:** Tall, sharp peak with heavy outliers.
- **Platykurtic:** Flat, wide peak.

4. Measure of Association: Correlation Coefficient

When analyzing bivariate data, we use the **Pearson Correlation Coefficient** (r) to measure the strength and direction of a linear relationship.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Key Properties of r :

- Always ranges between -1 and $+1$.
- $r = +1$ **or** -1 : Perfect linear relationship.
- $r = 0$: No linear relationship whatsoever.
- Measures *only* linear association, not non-linear curves!

The Scope of Our Data: Population vs. Sample

Before we can summarize data, we must understand where it comes from and what it represents. In statistics, we draw a strict line between two concepts:

The Population

The complete, exhaustive collection of all individuals, items, or events that we are interested in studying.

- The size of a finite population is typically denoted by uppercase **N**.
- *Example:* Every single registered student at the university.

The Sample

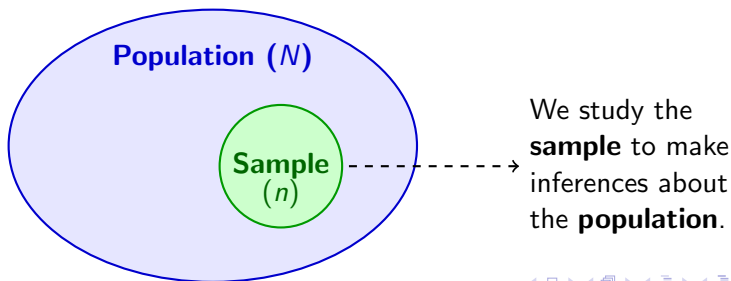
A strictly smaller subset of the population, selected for actual observation and measurement.

- The size of a sample is typically denoted by lowercase **n**.
- *Example:* A random group of 50 students from the university.

Why Do We Sample?

If we want to know about the population, why not just measure the whole population (a **census**)?

- **Impracticality:** Populations are often too large (e.g., all manufactured parts from a factory).
- **Cost and Time:** Measuring everyone is extremely expensive and slow.
- **Destructive Testing:** Sometimes measuring an item destroys it (e.g., testing the lifespan of a lightbulb or the safety of a car in a crash).



Where does the data come from? The Random Experiment

Data doesn't just appear; it is generated through observation or physical action. We call the mechanism that generates our data a **Random Experiment**.

Definition of a Random Experiment

A process, action, or observation that satisfies three specific conditions:

- 1 It can be **repeated** under virtually identical conditions.
- 2 The set of all possible outcomes is **known in advance**.
- 3 The exact outcome of any single trial **cannot be predicted with certainty** beforehand.

Examples of Random Experiments

A random experiment does not necessarily mean "mixing chemicals in a lab." It simply refers to any systematic observation with an uncertain result.

Example 1: Quality Control

Experiment: Selecting a motherboard off a manufacturing line and testing it.

Possible Outcomes: {Defective, Functioning}

Uncertainty: We know it will be one or the other, but we do not know which one until we test it.

Examples of Random Experiments

Example 2: Data Collection

Experiment: Asking a randomly selected incoming student their primary programming language.

Possible Outcomes: {Python, R, C++, Java, Julia, ...}

Uncertainty: The list of potential languages is known, but the specific student's answer is uncertain until recorded.

By repeating a random experiment n times, we generate our sample dataset of size n .

Defining the Outcome Space: The Sample Space

Definition (Sample Space)

The **sample space** of a random experiment, denoted by $\mathcal{S}$ (or the Greek letter Ω), is the set of all possible outcomes of that experiment. An individual result in this set is called an **outcome** or a **sample point**.

Classification of Sample Spaces

Depending on the nature of the experimental outcomes, $\mathcal{S}$ can be classified as:

- **Discrete (Finite or Countable):** Outcomes can be listed in a sequence (aligns with categorical or discrete numerical data).
- **Continuous (Uncountable):** Outcomes form an interval of real numbers, representing precise measurements.

Revisiting Our Previous Scenarios

Examples: Discrete vs. Continuous

- **Discrete (Categorical):** Testing a component off the manufacturing line for quality control:

$$\mathcal{S} = \{\text{Defective}, \text{Functioning}\}$$

- **Discrete (Numerical):** Counting the number of dot balls bowled in a single over of a T20 cricket match:

$$\mathcal{S} = \{0, 1, 2, 3, 4, 5, 6\}$$

- **Continuous (Numerical):** Measuring the exact midday temperature T (in $^{\circ}\text{C}$) at the factory:

$$\mathcal{S} = \{T \in \mathbb{R} \mid T \geq 18\}$$

Raw Data and Order Statistics

To define summary statistics formally, we first need to establish our notation for a dataset containing n observations.

Raw Data

Let the unordered observations in our dataset be denoted by:

$$x_1, x_2, x_3, \dots, x_n$$

where x_i represents the i -th recorded observation.

Order Statistics

Many statistics (like the median) require sorted data. When we sort the data in ascending order, we use parentheses in the subscript:

$$x_{(1)} \leq x_{(2)} \leq x_{(3)} \cdots \leq x_{(n)}$$

The Sample Mean

The **Sample Mean** ($\bar{x}$) is the arithmetic average of the data.

Definition

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Physical Interpretation:

- The mean acts as the **center of mass** of the dataset.
- If you place equal weights on a see-saw at positions corresponding to each x_i , the see-saw will perfectly balance exactly at $\bar{x}$.
- Consequently, the sum of all deviations from the mean is always zero:
 $\sum_{i=1}^n (x_i - \bar{x}) = 0$.

The Sample Median

The **Sample Median** (M) is the exact midpoint of the sorted data. Its formal definition depends on whether n is odd or even.

Definition using Order Statistics

$$M = \begin{cases} x_{(\frac{n+1}{2})} & \text{if } n \text{ is odd} \\ \frac{1}{2} \left(x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)} \right) & \text{if } n \text{ is even} \end{cases}$$

Example

- **Odd** ($n = 5$): 1, 3, **5**, 8, 9 $\implies M = x_{(3)} = 5$
- **Even** ($n = 6$): 1, 3, **5**, **7**, 8, 9 $\implies M = \frac{5+7}{2} = 6$

The Sample Mode

The **Sample Mode** is the value (or values) that appear most frequently in the dataset.

Definition

Let $f(v)$ represent the absolute frequency (count) of a distinct value v within our dataset. The mode is the value M_o such that:

$$f(M_o) \geq f(v) \quad \text{for all other values } v$$

Properties of the Mode:

- **Categorical Data:** It is the *only* measure of central tendency that applies to nominal data (e.g., finding the most popular programming language).
- **Non-Uniqueness:** A dataset can have one mode (unimodal), two modes (bimodal), or multiple modes (multimodal).
- **No Mode:** If every value appears exactly once (which is very common in raw continuous data), the dataset simply has no mode.

Mean vs. Median: The Problem of Outliers

Why do we need multiple measures of center? The answer lies in **robustness**, how resistant a statistic is to extreme values.

Dataset A: 40, 45, 50, 55, 60

- Mean ($\bar{x}$) = 50
- Median (M) = 50

Dataset B (One outlier added): 40, 45, 50, 55, 60, **1000**

- Mean ($\bar{x}$) = 208.3 (*Pulled drastically to the right*)
- Median (M) = 52.5 (*Remains highly stable*)

Key Takeaway

The mean utilizes the exact magnitude of every data point, making it sensitive to outliers. The median only cares about the *rank* or relative position, making it a robust statistic.

Generalizing the Median: Quartiles and Percentiles

If the median divides the dataset into two equal halves, we can extend this logic to divide the data into more segments.

Quartiles

Quartiles divide the sorted dataset into four equal parts (25% each):

- Q_1 (**First Quartile**): The median of the lower half (25th percentile).
- Q_2 (**Second Quartile**): The Median (50th percentile).
- Q_3 (**Third Quartile**): The median of the upper half (75th percentile).

The Interquartile Range (IQR):

$$\text{IQR} = Q_3 - Q_1$$

The IQR measures the spread of the middle 50% of the data. Because it ignores the lowest 25% and highest 25%, it is highly robust against outliers (unlike the total Range).

Variance and Standard Deviation

How do we mathematically capture how far, on average, the data points spread away from the mean?

Sample Variance (s^2)

The average of the *squared* differences from the mean:

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample Standard Deviation (s)

Because variance squares the original units, we take the square root to return to the original scale:

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

An Alternative Spread: Mean Absolute Deviation (MAD)

Definition of MAD

The Mean Absolute Deviation (MAD) is the arithmetic average of the absolute differences between each data point and the sample mean:

$$\text{MAD} = \frac{1}{n} \sum_{i=1}^n |x_i - \bar{x}|$$

MAD vs. Standard Deviation (s):

- **Intuition:** MAD literally represents the "average distance from the mean," which is how most people naturally think about spread.
- **Robustness:** Because standard deviation *squares* the deviations, it heavily penalizes extreme outliers. MAD treats all distances linearly, making it less sensitive to extreme values.
- **Mathematical limitation:** Absolute values suffer from differentiability issues.

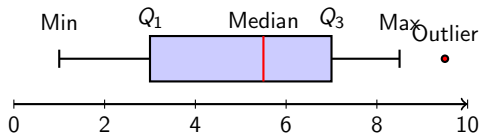
The Five-Number Summary & Boxplots

By combining robust order statistics, we can summarize the entire location, spread, and shape of a dataset into just five numbers.

Five-Number Summary:

- ① Minimum ($x_{(1)}$)
- ② First Quartile (Q_1)
- ③ Median (Q_2)
- ④ Third Quartile (Q_3)
- ⑤ Maximum ($x_{(n)}$)

The Boxplot (Box-and-Whisker)



The boxplot visually bridges order statistics with the concept of data dispersion.

Mathematical Moments: Raw Moments

In physics, "moments" describe the distribution of mass relative to an axis (like the center of mass or moment of inertia). In statistics, we use moments to describe the shape of a data distribution.

Raw Moments (Moments About the Origin)

The k -th raw sample moment, denoted as m'_k , is the average of the data values raised to the power of k :

$$m'_k = \frac{1}{n} \sum_{i=1}^n x_i^k$$

The First Raw Moment ($k = 1$):

$$m'_1 = \frac{1}{n} \sum_{i=1}^n x_i^1 = \bar{x}$$

Moments About an Arbitrary Point

Sometimes, analyzing data from the origin (0) is computationally inconvenient or lacks context. We can calculate moments relative to any arbitrary constant value A .

Moments About Point A

The k -th sample moment about an arbitrary point A is:

$$m_k(A) = \frac{1}{n} \sum_{i=1}^n (x_i - A)^k$$

- If $A = 0$, this collapses back into the raw moment (m'_k).
- This generalized form is often used as a computational stepping stone to derive other statistics without dealing with large numbers.

Central Moments (Moments About the Mean)

To understand the *spread* and *shape* of a dataset independent of where it is located on the number line, we set our arbitrary point A to the sample mean $\bar{x}$.

Central Moments

The k -th central sample moment, denoted as m_k , measures deviations around the sample mean:

$$m_k = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k$$

The First Four Central Moments

The first four central moments define the core characteristics of our dataset:

- **First Central Moment ($k = 1$):**

$$m_1 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) = 0$$

(The positive and negative deviations from the mean perfectly cancel out.)

- **Second Central Moment ($k = 2$):**

$$m_2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Higher-Order Central Moments: Skewness and Kurtosis

As we raise the power k , the equations become highly sensitive to data points that are far away from the mean (the tails).

- **Third Central Moment ($k = 3$): Skewness**

$$m_3 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3$$

Because 3 is an odd power, negative deviations remain negative. If $m_3 \neq 0$, the distribution is asymmetric (skewed).

- **Fourth Central Moment ($k = 4$): Kurtosis**

$$m_4 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4$$

Because 4 is an even power, everything becomes positive, but extreme outliers are magnified enormously. This measures the thickness/weight of the tails.

Effect of Transformation: Change of Origin (Shift)

What happens to our central moments if we add or subtract a constant value c from every data point? This is known as a **change of origin**.

Let our new dataset be defined as $y_i = x_i + c$.

The new mean shifts by the same constant: $\bar{y} = \bar{x} + c$

Now, let's calculate the k -th central moment for Y :

$$m_k(Y) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^k$$

$$m_k(Y) = \frac{1}{n} \sum_{i=1}^n [(x_i + c) - (\bar{x} + c)]^k$$

$$m_k(Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k = m_k(X)$$

Key Property: Shift Invariance

Central moments are unaffected by shifting the location of data.

Effect of Transformation: Change of Scale

What happens if we multiply every data point by a constant c (e.g., converting measurements from meters to centimeters)? This is a **change of scale**.

Let our new dataset be $y_i = c \cdot x_i$.

- The new mean scales by the same constant: $\bar{y} = c \cdot \bar{x}$

Calculating the k -th central moment for Y :

$$m_k(Y) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^k$$

$$m_k(Y) = \frac{1}{n} \sum_{i=1}^n (cx_i - c\bar{x})^k$$

$$m_k(Y) = \frac{1}{n} \sum_{i=1}^n c^k (x_i - \bar{x})^k = c^k \left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^k \right]$$

Scaling Central Moments: The General Rule

From our derivation, we extract the golden rule for scaling data:

$$m_k(Y) = c^k \cdot m_k(X)$$

Applying this to our known statistics:

- **Variance ($k = 2$):** If you multiply data by c , the variance is multiplied by c^2 . (This is why we take the square root to get Standard Deviation, which scales linearly by c).
- **Skewness Moment ($k = 3$):** Scales by c^3 .
- **Kurtosis Moment ($k = 4$):** Scales by c^4 .

General Linear Transformation

If we apply both operations simultaneously, $y_i = cx_i + d$, the constant d vanishes, and only the scaling factor c affects the central moment:

$$m_k(cX + d) = c^k \cdot m_k(X)$$

Minimizing the Second Moment

We established that the second central moment uses the mean ($A = \bar{x}$). But mathematically, why is the mean the optimal anchor point for measuring variance?

Let's find the value of A that minimizes the sum of squared deviations:

$$S(A) = \sum_{i=1}^n (x_i - A)^2$$

Algebraic Trick: We introduce the sample mean $\bar{x}$ by adding and subtracting it inside the parentheses:

$$S(A) = \sum_{i=1}^n (x_i - \bar{x} + \bar{x} - A)^2$$

Now, group the terms to prepare for binomial expansion $(a + b)^2$:

$$S(A) = \sum_{i=1}^n [(x_i - \bar{x}) + (\bar{x} - A)]^2$$

Minimizing the Second Moment

Expanding the square yields three components:

$$S(A) = \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^n 2(x_i - \bar{x})(\bar{x} - A) + \sum_{i=1}^n (\bar{x} - A)^2$$

Notice that $2(\bar{x} - A)$ is a constant with respect to the summation index i , so we can factor it out of the middle term:

$$S(A) = \sum_{i=1}^n (x_i - \bar{x})^2 + 2(\bar{x} - A) \sum_{i=1}^n (x_i - \bar{x}) + n(\bar{x} - A)^2$$

Recall our rule for the first central moment: $\sum_{i=1}^n (x_i - \bar{x}) = 0$. The entire middle term vanishes!

$$S(A) = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - A)^2$$

To minimize the total sum, we must force the second term to zero, which happens **strictly** when $A = \bar{x}$.

Symmetric Sample?

To understand how skewness works, we first need to define what it means for a dataset to be **symmetric**.

Algebraic Symmetry

A dataset is perfectly symmetric around its mean $\bar{x}$ if, for every data point located a distance d above the mean, there is a corresponding data point located the exact same distance d below the mean.

Let the deviation of any point from the mean be $d_i = x_i - \bar{x}$.

In a symmetric sample, if you have a positive deviation $(+d)$, there must exist a matching negative deviation $(-d)$ in the dataset.

Note: This is an unlikely scenario to happen. But something close to this can happen. In reality for a symmetric population a sample will probably not be perfectly symmetric but close. So m_3 will be close to 0.

Why $m_3 = 0$ for a Symmetric Sample

Now, let's calculate the third central moment (m_3) for this perfectly symmetric dataset:

$$m_3 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3$$

Replacing $(x_i - \bar{x})$ with our deviation notation d_i , the sum becomes:

$$m_3 = \frac{1}{n} \sum_{i=1}^n d_i^3$$

Because the dataset is symmetric, we can pair up every positive deviation ($+d$) with its corresponding negative deviation ($-d$). Let's look at what happens when we cube these pairs:

- The positive deviation cubed: $(+d)^3 = +d^3$
- The negative deviation cubed: $(-d)^3 = -d^3$

When we sum these paired terms, they perfectly cancel each other out:

$d^3 + (-d^3) = 0$. Since all pairs cancel, the entire sum collapses to zero.

The Need for scale-invariant summaries

We already proved that m_3 scales by c^3 and m_4 scales by c^4 . This creates a practical problem for comparing datasets:

- If we measure a dataset in meters, m_3 has units of meters³.
- If we measure the exact same data in centimeters, m_3 becomes 1,000,000 times larger!

The Solution: Standardization

To create universal measures of shape that allow us to compare completely different datasets (e.g., temperature in C vs F, different data with different units), we must make our moments **dimensionless**. We do this by dividing them by our measure of spread (variance/standard deviation) raised to a matching power.

Moment Coefficient of Skewness (g_1)

To create a dimensionless measure of skewness, we take the third central moment (m_3) and divide it by the second central moment (m_2) raised to the power of $3/2$.

Fisher-Pearson Coefficient of Skewness

$$g_1 = \frac{m_3}{(m_2)^{3/2}} = \frac{m_3}{s^3}$$

Interpretation:

- $g_1 = 0$: The data is perfectly symmetric.
- $g_1 > 0$: Right-skewed (positive skew). The right tail is longer.
- $g_1 < 0$: Left-skewed (negative skew). The left tail is longer.

Pearson's Empirical Skewness

Before moments were widely calculated by computers, statisticians used simpler algebraic formulas based on the geometry of the Mean, Median, and Mode.

Recall our rule for robustness: extreme values pull the Mean away from the Median and Mode.

Pearson's First Coefficient (Mode-based)

$$Sk_1 = \frac{\bar{x} - \text{Mode}}{s}$$

Pearson's Second Coefficient (Median-based)

Because the mode can be unstable or undefined, the median is preferred:

$$Sk_2 = \frac{3(\bar{x} - \text{Median})}{s}$$

Moment Coefficient of Kurtosis (g_2)

Kurtosis measures the "tailedness" (and conversely, the peakedness) of a dataset. To make the fourth central moment dimensionless, we divide it by the second central moment squared.

Coefficient of Kurtosis

$$g_2 = \frac{m_4}{(m_2)^2} = \frac{m_4}{s^4}$$

The Mathematical Benchmark: In statistics, the standard, perfectly symmetric "bell-shaped curve" evaluates to a kurtosis of exactly **3**. We use 3 as the baseline to compare all other datasets.

Excess Kurtosis and Classification

Because the standard bell curve has a kurtosis of 3, statistical software usually reports **Excess Kurtosis**:

$$\text{Excess Kurtosis} = g_2 - 3$$

Based on this, we classify data distributions into three shape categories:

- **Mesokurtic (Excess = 0)**: Matches the baseline bell shape.
- **Leptokurtic (Excess > 0)**: "Lepto" means thin/slender. The distribution has a sharper peak and **fatter, heavier tails** (meaning extreme outliers are more common than in a standard curve).
- **Platykurtic (Excess < 0)**: "Platy" means broad/flat. The distribution has a flatter peak and **thinner tails** (extreme outliers are rare).

Moment Inequalities: Cauchy-Schwarz

How do different moments relate to one another? They are not entirely independent; the values of lower moments mathematically restrict the possible values of higher moments.

The Cauchy-Schwarz Inequality (Sample Version)

For any two sets of real numbers a_i and b_i :

$$\left(\frac{1}{n} \sum_{i=1}^n a_i b_i \right)^2 \leq \left(\frac{1}{n} \sum_{i=1}^n a_i^2 \right) \left(\frac{1}{n} \sum_{i=1}^n b_i^2 \right)$$

Application to Raw Moments:

If we set $a_i = x_i$ and $b_i = 1$, the inequality collapses to:

$$(\bar{x})^2 \leq \frac{1}{n} \sum_{i=1}^n x_i^2 \implies (m'_1)^2 \leq m'_2$$

The Skewness-Kurtosis Bound

A more advanced application of Cauchy-Schwarz limits how skewed a distribution can be for a given kurtosis.

Let $a_i = (x_i - \bar{x})$ and $b_i = (x_i - \bar{x})^2$. Applying the inequality yields a fundamental relationship between the second, third, and fourth central moments:

$$(m_3)^2 \leq m_2 \cdot m_4$$

Dividing both sides by $(m_2)^3$ translates this boundary into our standardized coefficients of skewness (g_1) and kurtosis (g_2):

$$g_2 \geq g_1^2 + 1$$

This tells us that highly skewed data mathematically requires "fat tails" (high kurtosis).

The Absolute Lower Bound of Kurtosis

We can also use Cauchy-Schwarz to find the absolute minimum possible value for the moment coefficient of kurtosis (g_2).

Let $a_i = 1$ and $b_i = (x_i - \bar{x})^2$. Applying the Cauchy-Schwarz inequality:

$$\left(\frac{1}{n} \sum_{i=1}^n (1) \cdot (x_i - \bar{x})^2 \right)^2 \leq \left(\frac{1}{n} \sum_{i=1}^n 1^2 \right) \left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4 \right)$$

This elegantly simplifies to our central moments:

$$(m_2)^2 \leq (1) \cdot m_4$$

The Minimum Kurtosis

Dividing both sides by $(m_2)^2$ isolates our kurtosis coefficient:

$$g_2 = \frac{m_4}{m_2^2} \geq 1$$

Moving to Bivariate Data

So far, we have summarized a single variable (univariate data) using moments. Now, suppose we measure **two** variables simultaneously for each individual.

Bivariate Data

Let our dataset consist of n pairs of observations:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

Example: x_i is the number of hours a student studied, and y_i is their final exam score.

We already know how to find the mean $(\bar{x}, \bar{y})$ and variance (s_x^2, s_y^2) for X and Y individually. But how do we mathematically measure how they **move together**?

Sample Covariance

To measure the joint variability of two variables, we define their **covariance**. It is a direct bivariate generalization of variance.

Definition of Sample Covariance

$$s_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Interpretation of the Sign:

- $s_{xy} > 0$: Positive relationship. When x is above its mean, y tends to also be above its mean (they move in the same direction).
- $s_{xy} < 0$: Negative relationship. When x is above its mean, y tends to be below its mean (they move in opposite directions).
- *Note:* If $Y = X$, then $s_{xx} = \frac{1}{n} \sum (x_i - \bar{x})^2 = s_x^2$. Variance is simply the covariance of a variable with itself!

Pearson Correlation Coefficient (r)

We standardize the covariance by dividing it by the product of the individual sample standard deviations (s_x and s_y).

Sample Correlation Coefficient

$$r = \frac{s_{xy}}{s_x s_y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Mathematical Bounds (via Cauchy-Schwarz):

By applying the Cauchy-Schwarz inequality to the sequences $a_i = (x_i - \bar{x})$ and $b_i = (y_i - \bar{y})$, it mathematically guarantees that $(s_{xy})^2 \leq s_x^2 s_y^2$.

Dividing by the right side proves that correlation is strictly bounded:

$$-1 \leq r \leq 1$$

Interpreting the Correlation Coefficient

The correlation coefficient r is a **dimensionless** measure of the strength and direction of the **linear** relationship between two variables.

- $r = 1$: Perfect positive linear relationship (all data points lie exactly on an upward-sloping straight line).
- $r = -1$: Perfect negative linear relationship (all data points lie exactly on a downward-sloping straight line).
- $r = 0$: No *linear* relationship. The variables might be completely independent, or they might have a strong **non-linear** relationship (like a parabola) that r cannot detect.

Golden Rule: Correlation strictly measures linear association. It does not imply causation between X and Y .